

Thermal conductivity · PREFIX.kappa.h5

/

@ schema = alamode:kappa_result
@ format_version, alamode_version
@ alamode_git_commit, created_date
@ temperature_resolved = 0 / 1

/metadata

temperatures (nT,) [K] · fcs_file ()

classical () · ismear () · isotope ()
smearing_width () [cm⁻¹]
isotope_factors (nelem,)

/metadata/
PrimitiveCell

lattice_vector (3, 3) [bohr] · volume () [bohr³]

fractional_coordinate (natom, 3)
atomic_kinds (natom,) · elements (nelem,)
number_of_atoms () · number_of_elements ()

/metadata/
input_variables

@ input tags and their values

/metadata/
input_variables_runs/
280.00, ...
FC2_TEMPERATURE runs

@ input tags per temperature run

/metadata also has:
fc2_source () · fc2_temperatures (nT,) [K]

/scattering3ph

@ kmesh, nk_irred, nbranches

frequencies (nk_irred, nbranch) [cm⁻¹]
gamma (nk_irred*nbranch, nT) [cm⁻¹]
gamma_computed (nk_irred*nbranch,)
velocities (nk, nbranch, 3) [m/s]
velocity_diad (nk, nbranch, 3, 3) [(m/s)²]
xk_irred (nk_irred, 3) · weights (nk_irred,)
equiv_knum (nk,) · equiv_offsets (nk_irred+1,)

/scattering3ph

When temperature_resolved = 1:

frequencies (nT, nk_irred, nbranch) [cm⁻¹]
velocities (nT, nk, nbranch, 3) [m/s]
velocity_diad (nT, nk, nbranch, 3, 3) [(m/s)²]
gamma_computed (nk_irred*nbranch, nT)
gamma keeps the shape shown above

/scattering4ph
INCLUDE_4PH=1

Same layout as 3ph at temperature_resolved = 0,
except velocity_diad (not in the 4ph dump)
Own kmesh, nk_irred and nbranches

/scattering/isotope
ISOTOPE>0

gamma (nk_irred, nbranch) [cm⁻¹]

With temperature_resolved = 1:
gamma (nT, nk_irred, nbranch) [cm⁻¹]

/kappa

kappa_peierls (nT, 3, 3) [W/mK]

valid (nT,) · uint8 if temperature_resolved = 1
valid (1,) · uint8 otherwise
@ formulation, boundary_length
@ includes_4ph_scattering
@ includes_isotope_scattering
@ includes_boundary_scattering

/kappa
INCLUDE_4PH=1

kappa_3ph_only (nT, 3, 3) [W/mK]

@ scattering_processes = 3ph+isotope

/kappa
KAPPA_COHERENT>0

kappa_coherent (nT, 3, 3) [W/mK]

kappa_total (nT, 3, 3) [W/mK]

/kappa
KAPPA_SPEC=1

kappa_spec (n_e, nT, 3) [W/mK/cm⁻¹]

energy_axis (n_e,) [cm⁻¹]

/iterativebte
SOLVER=IBTE

Q (nT, nk_irred, nbranch)

dF (nT, nk_irred, nbranch, 3) [bohr/K]
kappa (nT, 3, 3) [W/mK]
computed (nT,) · converged (nT,)
@ kmesh, nk_irred, nbranches

Folder: group/path · box: datasets · note: attributes

Dashed: optional · @: attribute · (): scalar · []: stored unit

nT: temperatures · nk: full mesh · nk_irred: irreducible points

nbranch: branches · natom: cell atoms · nelem: elements · n_e: energy bins